

#DataForUkraine: Adapting Social Science Tools for Crisis Response

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The full-scale invasion of Ukraine by the Russian Federation shocked many scholars of the region, international observers, and policymakers. In the initial weeks and months of the invasion, relief organizations and government agencies lacked data about events on the ground and struggled to mount an effective response. The shocking extent of violence, mass flows of people, and lack of data collection infrastructure meant that there was an urgent need for new means of rapidly producing and disseminating data. At the request of policymakers, a team of Ukraine scholars and computational social scientists constructed a social media-based event detection system providing publicly available event data in near real-time. In this reflection, we demonstrate how social scientists can provide data to decision-makers during crises. We discuss the challenges involved and illustrate the data's utility for decision-making and academic research using case studies and descriptive analysis. We hope these insights can provide guidance for researchers and practitioners as new crises emerge around the world.

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How can social scientists use their technical and substantive knowledge to support decision-makers during major crises? When the Russian Federation invaded Ukraine on 24 February, 2022, this was a question many social scientists asked themselves. The full-scale invasion shocked many scholars, international observers, and policymakers, and in the initial weeks and months of the invasion, relief organizations and government agencies lacked data about events on the ground. The extent of the violence, the scale of internal displacement, and a lack of real-time information made clear the need for the agile production and dissemination of data to address a critical need.

At the request of policymakers at the United States Agency for International Development (USAID), a team of Ukraine-scholars and computational social scientists rapidly constructed a social media-based, near-realtime event detection system to provide publicly available data on developments in Ukraine. Combining deep country-expertise with computational tools, this system distills massive volumes of information into *hourly* geo-coded data updated in near-realtime that can both inform decision-making and answer important research questions.

In this reflection, we describe our approach to identifying reliable sources of information on social media, classifying text and geotagging across multiple languages, and disseminating these data to the public. We then describe important characteristics of the data, using case studies to

illustrate how events on the ground are reflected in our data and analyzing heterogeneity in information coming from different languages and user-communities. We also discuss policy-relevant applications ranging from providing immediate relief for refugees to future war crimes investigations and demonstrate how our data allows scholars to understand how information around events spread in realtime. Throughout, we focus on the general nature of the challenges we encountered and the solutions we implemented in the hopes that it may inform analogous data collection efforts in other contexts and on other platforms.

Social Media Data for Event Detection

In March 2022, USAID approached DevLab@Penn asking for data on the timing and location of events in Ukraine that might guide the international community's efforts to support the Ukrainian government and its citizens. Doing so required building a sustainable pipeline that could produce and disseminate up-to-date data on an ongoing basis at a relatively low-cost. To accomplish this, we turned to a popular social media platform: Twitter. The popularity of Twitter among journalists, activists, and policymakers means that the platform is often the first place where events are reported and discussed. Furthermore, Twitter's API provides easy access to rich information on the content of and engagement with tweets, providing a window into the

timing and perceived importance of events across communities of users, making it an increasingly important tool for researchers.

This process involved four primary tasks:

1. Building a sustainable pipeline for continually scraping and processing tweets
2. Identifying communities of users whose tweets are most likely to contain relevant information
3. Classifying tweets into topics of interest to policymakers and researchers
4. Extracting the locations of events reported in tweets

In the remainder of this section, we discuss how we implemented each of these tasks. Since these tasks would be inherent in any future crisis-driven data collection effort, we hope this discussion facilitates efforts by future research and practitioner teams.

First, building an infrastructure to scrape, store, and process large volumes of social media data required three components:

1. Backend collection with Twitter developer accounts using broad search terms in English, Russian, and Ukrainian
2. Parsing and storing data as digestible JSON (JavaScript Object Notation) file pieces

3. Reformatting and uploading tweets to a NoSQL (Not only SQL) Structured Query Language database

Initially, we employed a short list of broad keywords (including Russia, Ukraine, war, and occupation) to extract tweets that reference the conflict. This system was designed to collect the maximum amount of information relevant to the war. The volume of data quickly became enormous, measuring 5 million tweets per day in the early weeks of the invasion.

This enormous volume of data necessitated a strategy to identify and focus on the most relevant tweets. To filter the tweets, we identified a sample of Twitter accounts associated with Ukrainian ‘community leaders.’ Specifically, Ukraine scholars living in and outside of Ukraine and fluent in Ukrainian, Russian, and English manually selected more than 400 Twitter accounts belonging to individuals in Ukraine that hold important official positions, lead prominent organizations, or belong to investigative journalism networks.¹

We then employed community detection to map the network of influential accounts in which these leaders are embedded. After scraping the timeline (i.e. the entire twitter history) for each community leader, we created a network object of all retweets of their tweets and implemented a community detection algorithm (Leiden and walk trap community detection) to map the communities

in which these leaders are embedded. A primary connected network was then extended to a new round of high in-degree (i.e. influential) accounts, which gave us over 3,700 high-quality, well-connected Twitter accounts.

This community detection of augmented networks allows for the identification of high-quality accounts (high in-degree) connected to researcher-verified community leaders. In addition to reducing the volume of tweets to a manageable level, this approach uses the in-degree of accounts within our communities as a measure of reputation and credibility.

Looking at the identification of a sample of high in-degree users from each community and the content of their tweets and retweets, our Ukraine experts identified eight communities that were directly relevant to the conflict and categorized them by content. The communities were classified as: government-related; accounts related to liberal parties and some opposition; accounts related to the previous presidential administration; journalist and activist accounts; foreign accounts; and two smaller communities of activist accounts. These communities represent a diverse set of voices with varied agendas and relationships to the incumbent government. Figure 1 visualizes these eight core communities.

The third task was to identify categories of events (referred to as ‘event-types’) that could be useful to both policymakers and researchers. An initial list of events was

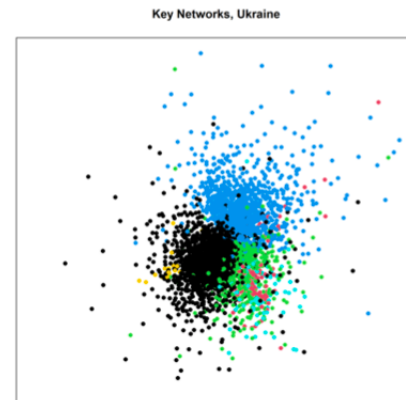


Figure 1

This graph shows the key network connectivity of the eight core communities.

produced by members of the research team with a substantive focus on conflict, social networks, civil society, or foreign aid and revised by the team’s Ukraine scholars. Finally, policymakers at USAID and several large, international NGOs provided feedback with a focus on their intended uses of the data. This process yielded four event-types: Humanitarian Support and Needs, Human Displacement, Human Rights Abuses, and Civilian Resistance.

To classify tweets according to these categories, we employed a list of keywords carefully adapted to the Ukrainian context. Because minimizing the time between a tweet being published and our ability to relay that information to the public was paramount, more computationally intensive approaches to text classification, such as the combination of machine translation and transformer-based large language models used in our other projects, was not

feasible.² Furthermore, it was important for our classification efforts to reflect the living languages local people used to describe events across our three languages.

For this reason, our team conducted extensive qualitative research of Twitter discourse to identify words and phrases being used to describe each event-type.³ Importantly, fluent or native speakers for each language ensured that we included relevant variations of words and phrases. In each case, keywords were validated by a minimum of three members of the team and then by two senior team leaders. Developing reliable keywords for each event-type was an iterative process with new keywords added as we learned about tweeting behavior around the conflict. When keywords generated too many ‘false positives’, they were replaced or eliminated from our list. After intensive testing against preliminary data, our final list contains roughly 600 keywords across the 4 event-types.

The first event-type, ‘Humanitarian Support and Needs,’ tracks events related to an ongoing humanitarian response or events that signaled a surge in the need for particular types of humanitarian assistance. Keywords include words and phrases describing a humanitarian response or describing specific forms of deprivation. An increase in the volume of tweets about ongoing or needed humanitarian support was expected to signal the timing and location of the need for both supplies and protection.

The second event-type, ‘Displacement,’ tracks events related to flows of refugees and internally displaced people.

Similar to Humanitarian Support and Needs, data on the timing and location of human displacements was expected to provide information about where and when assistance could/should be provided by external and domestic organizations to support these vulnerable individuals. Keywords focus on words and phrases that described displacement, attempts at evacuation (including when not successful as a result of blockades by the Russian military or due to violence), or the act of fleeing itself.

The third event-type, ‘Human Rights Abuses,’ tracks violations of basic human rights or specific international laws (such as the Geneva convention). Keywords include references to war crimes (such as deliberate targeting of civilians, kidnappings, mass killings, and torture, or the use of banned munitions) or human rights abuses (such as rape or repression of protests). Data on the timing and location of these events was expected to help in documenting where and when such actions occurred to support efforts to protect vulnerable populations or aid with future prosecutions.

The fourth event-type, ‘Civilian Resistance,’ tracks events related to civilian efforts to resist the occupation. Keywords include references to civilian collective action (such as protest events, the production of homemade weapons, or volunteer work) and data was expected to provide information on where and when resistance was occurring and to facilitate coordination. The inclusion of this category required significant discussion about any

potential implications for the safety of civilians engaging in resistance.

The final task required geo-locating events reported in tweets. Discussions with policymakers and collaborating NGOs made clear that event data would be most useful if they indicated where civilians were in need of support. However, geotagging tweets brings serious ethical challenges. Publishing data on the precise location of events may help guide humanitarian response but could also inform the Russian military or saboteurs seeking to block human flows, repress collective action, or cover-up evidence of crimes. Following discussions with Ukraine experts and team members on the ground, we limited the geographic granularity of our data to Ukraine's 136 *rayons*. Feedback suggested that this would provide sufficient detail to guide responses to regions with the greatest need while minimizing the risk of targeted blowback to specific communities.

Tweets come with location metadata capturing the location where each account was created. However, individuals have often moved since their account was created or are discussing events outside of the community where they live. To identify the rayon in which events being reported in tweets occurred, we search the text of each tweet and perform pattern matching to identify the name of any populated places being mentioned. Although this strategy provides more accurate location data, it entailed several challenges.

Due to numerous spellings of certain place names and variations caused by grammatical structure and conjugation across our three languages, this strategy required the construction of a database of names of all populated places in Ukraine with names from all three languages. To account for variation in spelling within and across languages, we used coordinates to match locations across three separate extant datasets listing over 40,000 unique place names. This was cross-checked by three different team members and corrected by Ukrainian native speakers. We then used the geo-location of each of these 40,000 places to identify which of Ukraine's contemporary rayons the populated place is located in (according to a shapefile obtained from collected from [OCHA \(2021\)](#)). This allows us to geolocate the content of tweets, so long as they use one of the spellings for places used by one of our three data sources.

Near-Realtime Data Dissemination

Our data production pipeline scrapes, stores, classifies, geotags, aggregates, and displays data publicly with an approximately 3-hour lag from reality.⁴ At 9pm in Kyiv, we have data capturing conditions through 6pm on the same day. However, these data are only useful for decision-makers if they are easily accessible. To disseminate our data to the public, we rely on interactive data visualizations using the R Shiny package and hosted on a public webpage.

To allow for easy triangulation by users on current conditions in Ukraine, we provide two sets of visualization

capturing both the timing and location of events. The first set provides an hourly time-series representing the share of all tweets scraped from our sample of accounts reporting on each event-type over the last 72 hours. As can be seen in Figure 2, a spike along the y-axis signals the potential occurrence of a major event in time.

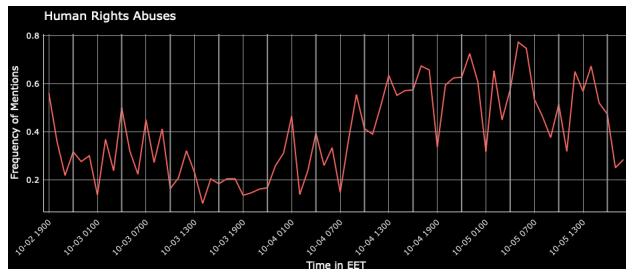


Figure 2
Hourly share of tweets reporting Human Rights Abuses over the last 72 hours (October 2022).

The second set of visualizations provide maps locating events in space. Figure 3 is a dynamic rayon-level map showing the share of tweets reporting on a user-determined each event-type for a user-determined time period within the last 72 hours (shown as the size of a point located in the rayon centroid) using the Leaflet R package (Cheng et al. 2022). Figure 4 is one of four static rayon-level maps showing the change in the share of tweets reporting on each event-type between the most recent two-hour period and the previous two-hour period. Figure 4 depicts the same information from Figure 2, but allows users to identify the rayon in which the event/s occurred.

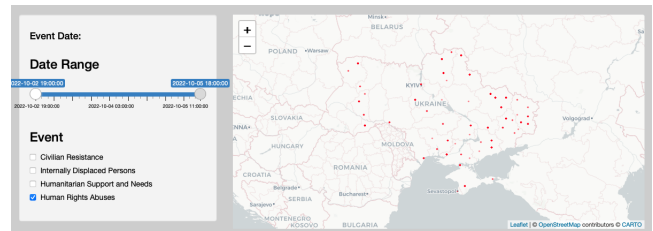


Figure 3
Rayon-level share of tweets reporting Human Rights Abuses over the past 72 hours (October 2022).

These visualization tools are accompanied by periodic reports produced by the research team. Reports were designed to monitor data quality, build trust among viewers, and provide opportunities to learn from the data. On a semi-weekly basis, the team extracts the 100 most retweeted tweets for each event-type in each of our three languages and reviews their content, how they were classified in our system, and what significance they have for understanding the situation faced by Ukrainian civilians.⁵ In the next section, we discuss what we learned from these reports and present results from additional analysis motivated by these reviews.

Validation and Analysis

In this section, we discuss results from periodic reviews of the data. These exercises confirm that the data capture the kinds of information that we sought to produce and disseminate to decision-makers. When taken together, these visualizations can provide an "early alarm" of major

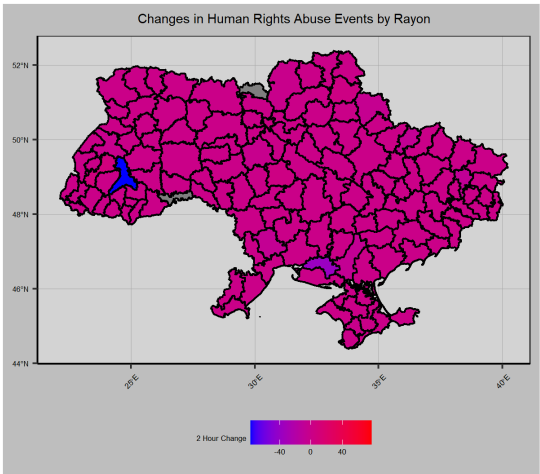


Figure 4
Rayon-level change in the share of tweets reporting Human Rights Abuses between the most recent two-hour period and the previous two-hour period.

Human Right Abuses as they occur but before they are reported on in the mainstream press. The data not only reliably capture major events, but our manual inspection of the data has identified a consistent pattern of spikes (identification of the occurrence of a major event in a given category) occurring 3-5 hours prior to related events being reported in the media. For example, in our analyses of events captured by our data on human rights abuses, we find that Twitter accounts report on Russian retreats and associated human rights abuses hours before official announcements or reporting.

Case Studies

In this section, we summarize information from reports produced by identifying major spikes in our data and linking these to specific events on the ground. To illustrate the value of the granular information, we consider a series of spikes occurring over the first two weeks of May 2022. By accessing the underlying tweets, we demonstrate that these spikes captured real and important events later documented by investigative journalism.

Specifically, spikes in our data were driven by reports of mass atrocities and war crimes taking place in Kherson, including reports on the imprisonment and replacement of local councils and heads of Kherson oblast, the torture of a Ukrainian Orthodox priest in Kherson, the shooting of children and elderly in the region, and the export of grain by Russian forces from Kherson to Crimea. Reports in Russian-language tweets also focused on human rights abuses in specific localities in Luhansk oblast. There were many tweets referring to rape being used as a weapon of war, looting, and Russian bombardments of civilian targets around Kharkiv and other eastern towns. Tweets also included information on the policies of Russian occupying forces such as efforts to restore normal transportation in Mariupol, plans to restore statues in Kharkiv and allegedly forcing students in the occupied territories to give blood for Russian soldiers.

Starting in May, there were also spikes in tweeting about war crimes in Bucha; our Ukrainian language corpus for May 8–9 provided tens of thousands of tweets covering six clusters of atrocities in Kharkiv, Sumy, Luhansk, Donetsk, and Zaporizhzhia as well as the continued targeting of civilians and civilian infrastructure in Borova, offensives in Barvinkove and Izyum in Kharkiv region, Odesa (repeatedly), Hannivka in Kryvyi Rih and Zelenodolsk in Dnipropetrovsk oblast, and in Mykolayiv.

Other examples include reports on April 24 of major events of human rights abuses occurring in Kharkiv, Kherson, Donetsk, and Luhansk Oblasts. Forced mobilization of young people and doctors is reported in Kharkiv, Kherson and Zaporizhzhia Oblasts. The largest event that occurred on April 24 was related to civilian resistance, involving a large auto convoy protest of citizens standing up against plans for an illegal referendum by ‘exiting’ the city. Reports indicated that Kherson continued to see excessive violence against peaceful civic resistance, with protesters regularly being shot at, tortured, imprisoned, and killed. Remarkably, realtime reporting on events in Kherson by locals continued to appear in our data even after the occupiers cut internet access.

Reviewing the top 100 retweeted Ukrainian-language tweets from July, August, and September 2022, we see similar evidence of spikes in reporting on our event categories providing an early warning of major events. The most active event-type in our data throughout this period

is Human Right Abuses, concentrated in Kharkiv, Zaporizhzhia, Kherson, Luhansk, and Donetsk - all partly occupied by the Russian Military and four of which were recently illegally annexed by Russia. The data show consistent reporting of human rights abuses and war crimes in territories close to the contact line or currently occupied by the Russian military in the southeast of Ukraine.

In these reports, an investigation of the underlying tweets driving major spikes in our event-types reveals two important patterns for future investigation. The first concerns the type of human rights abuses driving major spikes. These can be divided into three themes: 1) civilian residences being occupied/looted, attacks on civilian infrastructure, looting (agricultural or historical entities); 2) kidnapping, torture, indiscriminate killing of civilians, rape, repression, or intimidation of civilians; 3) weapons-use banned by the Geneva convention and targeted shelling resulting in nuclear threat.

The second pattern is that these reports seem to heighten in periods immediately preceding de-occupation and in periods directly following the de-occupation of a territory. The latter pattern is expected because once a territory is de-occupied, more information becomes available about the extent of crimes perpetrated during the occupation. However, the former pattern is surprising, suggesting that as Russian forces begin to lose control of territory, they increase the frequency or severity of human rights abuses. In a more recent analysis of these data, we find evidence

for the disproportionate targeting of educational settings, teachers, and school supplies (such as the destruction of children’s books) coupled with the forced destruction of Ukrainian passports. This is evidence of an attempt to attack Ukrainian civic identity and match up with other investigative and official reporting.

Language and Communities

Our in-depth analysis of the most retweeted tweets also reveals interesting heterogeneity in event-reporting across languages and user communities. First, there are interesting differences in the nature of the information conveyed in Ukrainian, Russian, and English-language tweets. Overall, the language that most closely reflects current events on the ground is Ukrainian. Not surprisingly, Russian language tweets are also useful for events in localities such as those in Luhansk and Donetsk Oblasts, though, even there, they tend to relate to events that are already at least a few days old (for the most shared ones at least). The most shared tweets in English tend to focus on events outside of Ukraine itself, whether declarations by international actors or responses to previous events in Ukraine.

Extending the analysis of the most popular tweets, we investigated the distinct patterns of top tweets across our eight user-communities. The top 20 retweeted posts for each community are in Ukrainian and English; almost none are in Russian. Across communities, English-language tweets reach a much broader audience and are

re-tweeted at higher rates, but Ukrainian-language tweets provide more substantive information on the nature of important events. Surprisingly, we find very little overlap in the substantive content of the information shared across different communities. This underscores our finding that not only can we identify different informational networks by focusing on the language of the tweet, but also different communities of actors develop distinct information networks.

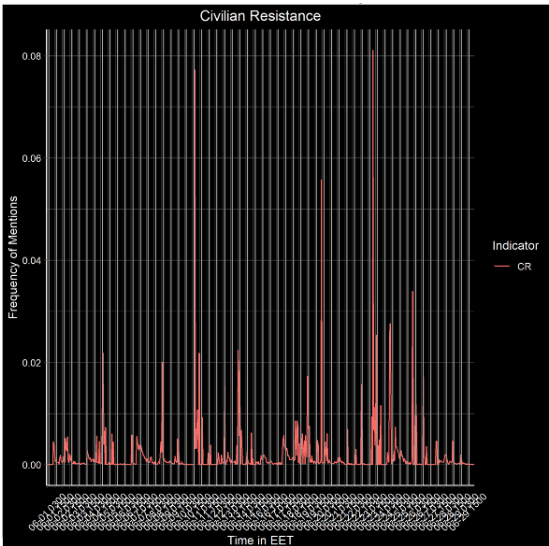


Figure 5
Change in the Frequency of Mentions of Civilian Resistance (June 2022).

The flow of information by community and language also reveals how audience considerations affect posting decisions. For example, consider the community of Ukrainian government officials and regional allies

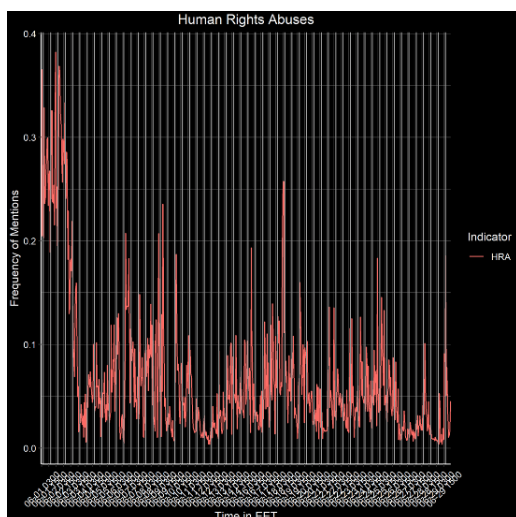


Figure 6
Change in the Frequency of Mentions of Human Rights Abuse (June 2022).

(232 accounts identified in one of the most active sub-networks), which posted a total of 298,901 tweets in the last 14 months. Figures 7 and 8 show the mean time-between-tweets in the seven months before and after the invasion for English and Ukrainian-language tweets, respectively. We visualise the mean time-between-tweets clustered into 100-time bins. Time-between-tweets is a common metric of the intensity of tweeting activity, with less time indicating more intensive activity.

In the seven months that preceded the invasion, Ukrainian government officials were already increasing the share and intensity of tweets posted in English as indicated by a decline in the time elapsed between each tweet. At the time of the invasion, the time elapsed between

tweets reaches its minimum (indicating its highest activity level), which is sustained for many months. In contrast, Ukrainian language tweets were published at a constant rate before the invasion. While the time between tweets declined with the beginning of the invasion, it returns to pre-invasion levels more quickly than English-language tweets.

Together these figures suggest that Ukrainian officials used English-language tweets to amplify information about the war for the consumption of the international community. Indeed, it is worth noting that the total volume of English tweets is almost 8 times higher than in Ukrainian.

Lessons on Interpretation & Public Use

A critical component of our public engagement has been engaging with users of our data about how to correctly interpret the outputs. For example, in a recent report, we take the example of the graphic output measuring Civilian Resistance for June 2022. A quick glance at Figure 5 might be interpreted as saying that around 10 June, 20 June, and 23 June our data captured three major events of civilian resistance. Similarly, Figure 6 we might be interpreted as showing that on 1-3 June, 9 June, 18 June, and 24 June our data collection captured four clusters of major events.

While both graphs show that events were reported, a close look at the Y-axis reveals that the magnitude of change in the frequency of mentions is very different in these two

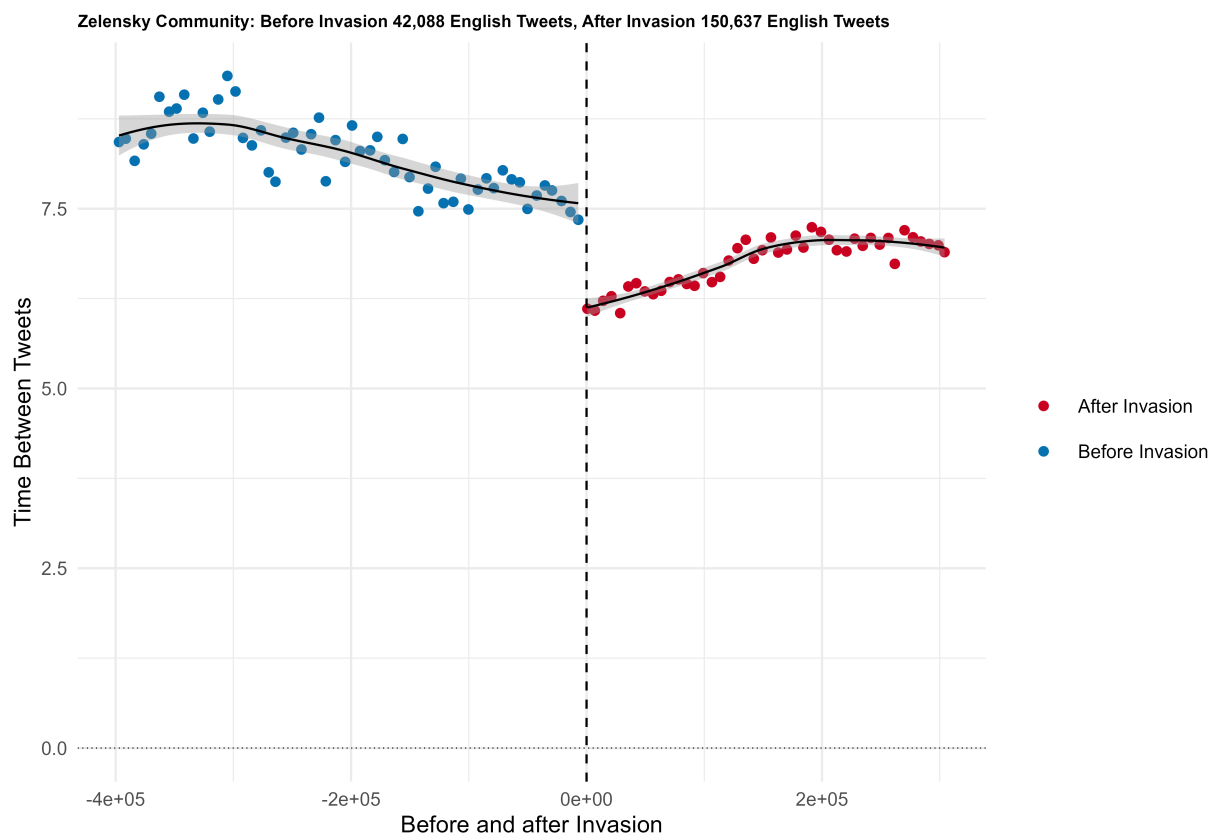


Figure 7

This graph describes the time elapsed between English-language tweets..

cases. For civilian resistance, the largest spike on 23 June corresponds to just over a 0.08 increase in the frequency of reporting, whilst the 1 June spike in the case of the Human Rights Abuses category corresponds to a near 0.4 increase in the frequency of reporting on the topic. Thus, we are clearly seeing major events occurring in the reporting of Human Rights Abuses, while the movement in our Civilian Resistance is capturing minor events.

Moreover, when we look at Figure 5 we see that there is constant clustering of reports. The baseline of Human Rights reports is therefore higher for the month of June

than it is in the case of Civilian Resistance. A 0.4 increase in the frequency of reports from a position of 0.3 frequency rate already is "higher" as a baseline than an increase from 0.0 to 0.08.

Finally, when we look at the maps depicting these events and their reporting for the month of June, we clearly see the difference in the magnitude of events reported, with larger circles depicting a larger magnitude of change in reporting – and hence suggesting a major Human Rights Abuse event has occurred in all of Kharkiv, Zaporizhzhia, Donetsk and Luhansk oblasts (please see Figure 10). Figure 9 also

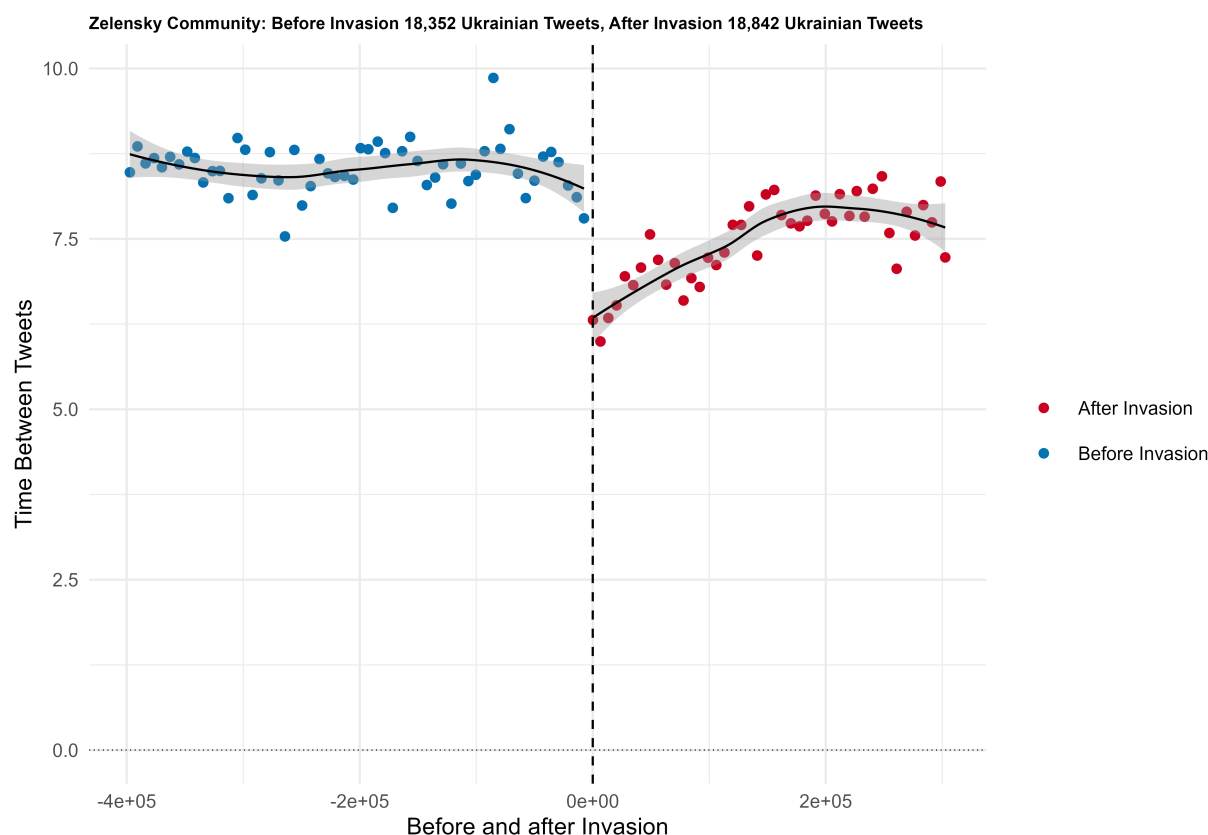


Figure 8

This graph describes the time elapsed between Ukrainian-language tweets..

shows that there were some reports of civilian resistance, but that the magnitude of reports on these events is much smaller. Again, here we see that there are more reports of civilian resistance in the occupied territories and in those territories closer to the contact line.

While this case study serves to guide data users, it also corresponds to the major findings of the #DataForUkraine project since May, showing that the same pattern has continued in June. We see an escalation of reports of Human Rights abuses; our first assessment suggests that there are

two waves of reports on Human Rights abuses, an escalation when the position of the Russian army in occupied territories is weakened or unclear and then a barrage of reports of abuses when these territories are de-occupied by the Ukrainian Army and the extent of events is evident.

Practical Applications and Policy Outreach

This article demonstrates how tools being developed and deployed by political scientists can provide valuable data to practitioners and policymakers in moments of crisis, when information is often scarce. Importantly, the rapid deployment of these tools provides opportunities for political

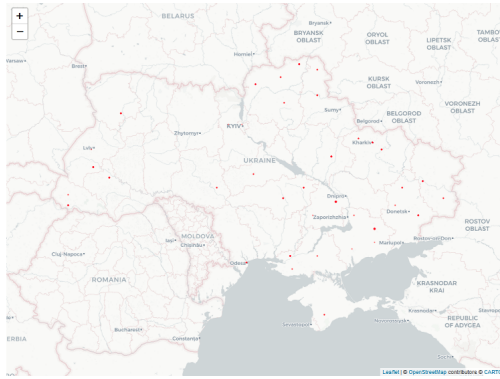


Figure 9
The Geographical Location and Magnitude of Events of Civilian Resistance Events, June 2022.

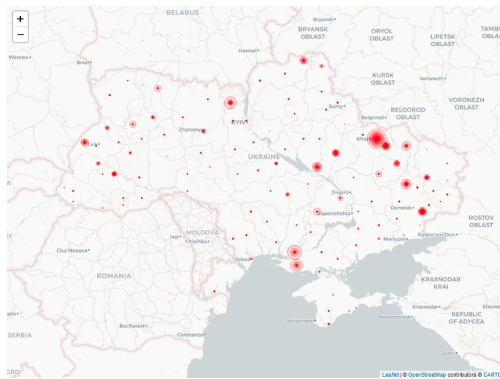


Figure 10
The Geographical Location and Magnitude of Events of Human Rights Abuse Events, June 2022.

scientists to expand their public engagement and provide an important service but doing so relies on deep collaboration between country experts and computational social scientists.

The initial motivation behind this project was to provide government agencies and non-governmental organizations with realtime data to inform emergency response. In the

initial stages, we briefed the Government of Ukraine, representatives of UK ministries, the FCDO in the UK, and the RAIO in the US, among other international NGOs and IGOs, on the data platform and its capabilities. Our team has also engaged the media and presented the project, its uses, and its initial findings in public-facing venues (like major policy focus workshops - one run by the British Academy, with involvement of the UK Ministry of Defence and Cabinet office representatives, Twitter spaces engaging leading journalists in Ukraine and the region, and public lectures to researchers on the region and others). We also regularly post “how to use” clips on our social media to ensure further usage of the system by ordinary citizens in Ukraine or internationally. Nevertheless, the challenge of getting the data into the hands of people who can effectively use it remains. Under conditions of war, it has hard to evaluate what the practical consequences of the project so far have been.

As the project has developed, the further benefits of a massive, searchable archive of tweets has become apparent. As an example, researchers studying the use of sexual violence as a tool of war are comparing qualitative evidence collected from survivor interviews with the #DataForUkraine archive have been able to extend their list of cases and look for patterns not apparent in the qualitative interviews.

Our data will also be a valuable resource to those working on other human rights violations and war crimes too.

The aggregated data and the underlying tweets can be used to identify the occurrence, frequency, and location of such acts and can be made available to help policymakers press for special tribunals and pursue justice for victims. Tweets may also be a resource for identifying individuals that can provide first person accounts of events that will be essential to prosecutors and scholars alike.

Finally, the basic approach adopted here could be adapted to other crises and other social media platforms. The core innovation is the creation of event data in near real-time in contexts where access to traditional media and other sources of reporting is difficult. The point is to find a way to collect and store crowd-sourced data. We believe that the possibilities for extensions are extremely broad. Any event that people tweet about and for which a clear and consistent set of regularly used keywords can be created could be processed and presented using the technique we have outlined here.

One possibility, is to use this model to collect similar event data on environmental challenges and patterns of response, including migration, protest, and other kinds of collective action. Existing databases of natural disasters are strong on the collection of environmental data with high human casualties but much weaker on smaller events and essentially silent on the social responses. The approach outlined here holds promise for the development of much richer datasets to guide both research and policy.

Notes

- 1 The authenticity of each account was validated with multiple cross-checks, including an overview by the team leader.
- 2 For a more computationally intensive approach that relies on more modern NLP models, see DevLab@Penn's [Machine Learning for Peace](#) website.
- 3 This research presented an ethical challenge, as reviewing of Twitter discourse risked exposing researchers in Ukraine to additional emotional trauma. To mitigate this, fluent speakers outside of Ukraine shared responsibility with Ukraine-based team members and assumed responsibility for the most disturbing subject-matter.
- 4 This includes an approximate 1-hour delay between when someone tweets and when it appears on the Twitter API.
- 5 The original intention had been to provide biweekly reports. However, the process of compiling these reports proved to be too labor-intensive given our limited resources. We now produce monthly and bi-monthly reports.

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Ethical Standards. This project received ethics approval from the University of Maryland and the University of Manchester.

Supplementary Material. The project website and interactive data dashboards can be found at this link: <https://web.sas.upenn.edu/mlp-devlab/ukraine-data/data-for-ukraine/>